Statistical Learning and Data Mining

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Financial Data Analysis: Predictive Analytics for Trading Actions

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Medical Imaging and Applications

Introduction

Objective: The project aims to utilize financial indicators to predict trading actions—buy, sell, or hold—using advanced machine learning techniques.

Significance: Accurate predictions can significantly impact trading strategies, enhancing profitability and risk management.

Data Exploration

- Training Data
  - 119 features, including 1 output
  - 8000 instances
- Testing Data
  - 119 features
  - 2000 instances
- Task: Whether to buy, sell or retain a stock based on values -1, 1, and 0 using key financial indicators which is a classification task
- Link: https://knowledgepit.ai/fedcsis-2024-challenge/

Data Analysis and Preprocessing

Data Loading and Transformation: Data is loaded from a CSV file and transformed by replacing commas with periods to standardize numerical formats, ensuring accurate data parsing.

Handling Missing Values: We identify 'NA' as missing data and empty strings as non-applicable entries. Columns with over 5% missing data are dropped to maintain data quality.

Feature Encoding: The 'Group' column is one-hot encoded to convert categorical into numerical data, facilitating machine learning processing.

Feature Engineering and Selection

Outlier Detection: Using the Interquartile Range (IQR), outliers are filtered from each feature based on calculated thresholds to reduce noise and improve model predictions.

Feature Selection via Correlation: Features strongly correlated with the 'Class' label are identified through correlation analysis. The top positive and negative correlations are selected for model training to maximize predictive power.

Model Training

Handling Imbalanced Data: SMOTETomek, a combination of Synthetic Minority Over-sampling Technique (SMOTE) and Tomek Links under-sampling, is used to balance the dataset, improving the model's generalizability.

Model Setup and Hyperparameter Tuning: A Gradient Boosting Classifier is configured within a pipeline. GridSearchCV is employed to optimize hyperparameters such as the number of trees, learning rate, and tree depth based on cross-validation scores.

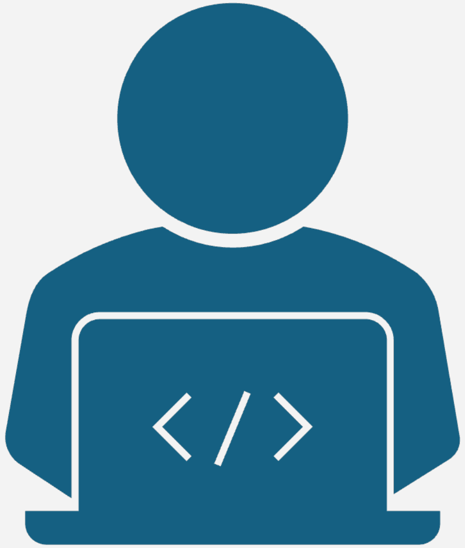
Model Evaluation

- **Evaluation Metrics**: Model performance is evaluated using accuracy metrics, a confusion matrix, and a detailed classification report to assess precision, recall, and F1 scores across classes.
- **Custom Cost Matrix:** A cost matrix is applied to the confusion matrix to quantify the financial impact of different types of prediction errors, providing insights into potential financial losses or gains..

Testing and Predictions

Test Data Preparation: Consistent preprocessing is applied to the test data, including imputation and encoding, ensuring that the data format matches the training dataset.

Generating Predictions: The trained model is used to create predictions for the test data. Predictions are then formatted according to the competition's requirements for submission.



Conclusion

The created predictive model generated encouraging results, indicating substantial potential for assisting with trading choices. Continued development and integration of larger data sets and advanced algorithms can potentially convert this model into an invaluable tool for traders, assisting them in maximizing gains and minimizing risks.

THANK YOU